A half-power vanishing-noise estimate for a quadratic kinetic mean-field game

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Abstract

We consider a variational kinetic mean-field game with quadratic acceleration cost, local quadratic congestion, and quadratic terminal congestion on $\mathbb{T}^d \times \mathbb{R}^d$. For a smooth, compactly supported probability datum, the density minimizers at velocity diffusivity ν and at zero diffusivity are at distance $O(\nu^{1/2})$ in spacetime L^2 and terminal L^2, in every dimension. The proof compares the two minimum values using kinetic Gaussian averaging and an entropy identity for the current flux. A positive convolution commuting with kinetic transport justifies this identity in the original weak admissible class, including the noncompact velocity domain and possible vacuum.

1 The variational problem and the estimate

Let $d \geq 1$, $T > 0$, and

$$Q = \mathbb{T}^d \times \mathbb{R}^d, \qquad S = (0,T) \times Q, \qquad D = \partial_t + v \cdot \nabla_x.$$

We use the unit-period torus with volume-one Haar measure. Fix a smooth, nonnegative probability density m_{in} on Q, compactly supported in velocity. For $\nu \in [0,1]$, consider the minimum of

$$\mathcal{J}(m,a) = \mathcal{A}(m,a) + \mathcal{G}(m), \qquad \mathcal{A}(m,a) = \int_S \frac{|a|^2}{2m}, \qquad \mathcal{G}(m) = \frac{1}{2}\int_S m^2 + \frac{1}{2}\int_Q m(T)^2, \tag{1}$$

under

$$Dm - \nu\Delta_v m + \operatorname{div}_v a = 0, \qquad m(0) = m_{\mathrm{in}}. \tag{2}$$

The acceleration flux a takes values in $\mathbb{R}^d$. At vacuum, $|a|^2/m$ is zero when $a = 0$ and infinite otherwise. The density $m(t)$ is a narrowly continuous curve of probability densities, has finite second velocity moment at every time, and satisfies (2) distributionally; the cost (1) is finite. In particular, the endpoint densities in (1) are the actual narrow traces. The weak equation, including these traces, reads

$$\int_Q m(T)\phi(T) - \int_Q m_{\mathrm{in}}\phi(0) = \int_S \left[m(\partial_t\phi + v\cdot\nabla_x\phi + \nu\Delta_v\phi) + a\cdot\nabla_v\phi\right] \tag{3}$$

for smooth tests periodic in x and compactly supported in v. We write (m_ν, a_ν) for minimizing pairs in this class and $E_\nu = \mathcal{J}(m_\nu, a_\nu)$. All conclusions below hold for any such pairs.

Vanishing-viscosity rates for kinetic mean-field games are raised by Tang and Zhang [1, Section 7, item 4(b)]. First-order kinetic games with local couplings have a variational theory on the noncompact velocity domain [2, Abstract]. Here we treat the specific quadratic problem (1)–(2). The comparison uses the entropy and current-velocity action identity appearing in the elliptic setting in Benamou, Carlier, Di Marino, and Nenna [3, eqs. (2.9)–(2.10)]. We derive its kinetic version directly, without assuming differentiability or Fisher information of an optimizer.

Theorem 1. *For every integer $d \geq 1$, every $T > 0$, and every smooth, nonnegative, compactly supported probability density m_{in} on Q, the minimizing densities defined above satisfy*

$$\|m_\nu - m_0\|^2_{L^2(S)} + \|m_\nu(T) - m_0(T)\|^2_{L^2(Q)} \leq 4L\nu \qquad (0 < \nu \leq 1). \tag{4}$$

Here the finite constant L, independent of ν, can be chosen as

$$\begin{aligned} B &= \frac{T+1}{2}\|m_{\mathrm{in}}\|^2_{L^2(Q)}, \qquad M_{\mathrm{in}} = \int_Q |v|^2 m_{\mathrm{in}}, \\ K &= e^T(M_{\mathrm{in}} + 2B + 2dT), \\ L &= |H_{\mathrm{in}}| + \frac{d}{2}\log(2\pi) + \frac{K+d}{2} + 1, \qquad H_{\mathrm{in}} = \int_Q m_{\mathrm{in}} \log m_{\mathrm{in}}. \end{aligned} \tag{5}$$

We use $0 \log 0 = 0$. In particular, $\|m_\nu - m_0\|_{L^2(S)} \leq 2\sqrt{L}\,\nu^{1/2}$.

The theorem is an estimate for the stipulated minimizing pairs. No existence theorem for a classical value function is needed. The exponent is independent of dimension; we do not assert its optimality.

Kinetic Gaussian averaging turns an inviscid pair into a noisy pair while decreasing its cost. For the reverse comparison, the entropy identity produces an inviscid competitor with the noisy minimizer's density and an excess cost bounded by $L\nu$. We justify this construction by positive convolution and restore the original datum before using minimality. Strict convexity of the two density costs then bounds the squared L^2 distance by the same order.

2 Moments and the noisy comparison

Averaging a finite-action pair against a probability measure of translations does not increase its action or either density-square cost. Indeed, weighted Cauchy–Schwarz gives

$$\frac{|\overline{a}|^2}{\overline{m}} \leq \frac{\overline{|a|^2}}{m}, \qquad \overline{m}^{\,2} \leq \overline{m^2}. \tag{6}$$

If $\overline{m} = 0$, nonnegativity and finite action imply $\overline{a} = 0$, so the stated vacuum convention is preserved. Also, finite action and total spacetime mass T give

$$\int_S |a| \leq \sqrt{2T\mathcal{A}(m,a)}. \tag{7}$$

Lemma 1. *Every admissible pair with $0 \leq \nu \leq 1$ satisfies*

$$\sup_{0\leq t\leq T} \int_Q |v|^2 m(t) \leq e^T \left(M_{\mathrm{in}} + 2\mathcal{A}(m,a) + 2dT\right).$$

Proof. Put $\psi_R(v) = |v|^2/(1+|v|^2/R^2)$ and $Y_R(t) = \int_Q \psi_R m(t)$. Direct differentiation gives $|\nabla\psi_R|^2 \leq 4\psi_R$ and

$$\Delta\psi_R = \frac{2d}{(1+|v|^2/R^2)^2} - \frac{8|v|^2/R^2}{(1+|v|^2/R^2)^3} \leq 2d.$$

For fixed R, use ψ_R in (3) after multiplication by a velocity cutoff at radius r, and then let $r \to \infty$. The extra flux term is bounded by $CR^2 r^{-1} \int_{\{|v|\geq r\}} |a|$, which vanishes by (7). The diffusion cutoff terms are bounded by a constant depending on R times r^{-1} or r^{-2} times the integrated

mass. Endpoint terms pass by bounded convergence. Thus Y_R is absolutely continuous, with its representative fixed by narrow continuity, and

$$Y_R' \le 2\sqrt{Y_R}\left(\int_Q \frac{|a|^2}{m}\right)^{1/2} + 2d \le Y_R + \int_Q \frac{|a|^2}{m} + 2d \quad \text{a.e.}$$

Integrating after multiplication by e^{-t} gives the claimed bound for $Y_R(t)$. Finally $\psi_R \uparrow |v|^2$, so monotone convergence proves the bound at every time. This argument does not require a previously known time-integrable second moment. □

For $\nu > 0$ and $t > 0$, define the probability Gaussian on $\mathbb{R}^{2d}$

$$k_{\nu,t}(y,w) = \left(\frac{\sqrt{3}}{2\pi\nu t^2}\right)^d \exp\left(-\frac{|w|^2}{4\nu t} - \frac{3|y - tw/2|^2}{\nu t^3}\right). \tag{8}$$

Its covariance blocks are $\mathrm{Cov}(y) = 2\nu t^3 I_d/3$, $\mathrm{Cov}(y,w) = \nu t^2 I_d$, and $\mathrm{Cov}(w) = 2\nu t I_d$. These follow by first integrating the conditional Gaussian in y, of mean $tw/2$ and covariance $\nu t^3 I_d/6$, then the Gaussian in w. Direct differentiation shows

$$\partial_t k_{\nu,t} + w \cdot \nabla_y k_{\nu,t} - \nu \Delta_w k_{\nu,t} = 0.$$

Periodize y on the torus, and denote convolution with this kernel by $C_\nu(t)$, with $C_\nu(0)$ the identity. Integration by parts in the shift variables yields

$$D(C_\nu(t) f) = C_\nu(t)(Df) + \nu \Delta_v (C_\nu(t) f). \tag{9}$$

The identity holds distributionally as well: apply the adjoint convolution to a smooth compact-velocity test and use the kernel equation. Cutting off this adjoint test is justified by its Gaussian tails, Lemma 1, and (7). The covariance tends to zero at $t = 0$, so the original initial trace is preserved.

The free-transport density $g(t,x,v) = m_{\mathrm{in}}(x - tv, v)$ has constant L^2 norm and zero flux. Equation (9) makes $(C_\nu(t)g, 0)$ an admissible noisy competitor. Together with (6), this proves

$$0 \le E_\nu \le B \quad (0 \le \nu \le 1), \qquad E_\nu \le E_0. \tag{10}$$

For the second inequality, use $(C_\nu(t)m_0, C_\nu(t)a_0)$ instead. The averaged density is a narrowly continuous probability curve and has velocity moment $\int_Q |v|^2 m_0(t) + 2\nu dt$. Narrow continuity follows by testing against bounded continuous functions: the original curve is uniformly tight on $[0,T]$, and the Gaussian shift law varies continuously, including its point-mass limit at zero. All terms of the cost, including the terminal term, decrease under this averaging. Lemma 1 and (10) give the uniform bound

$$\sup_{0 \le t \le T} \int_Q |v|^2 m_\nu(t) \le K \qquad (0 \le \nu \le 1), \tag{11}$$

where B and K are defined in (5).

For later use, set $H(f) = \int_Q f \log f$ for probability densities of finite absolute entropy. If $f \in L^2(Q)$ and $M(f) = \int_Q |v|^2 f < \infty$, this entropy is absolutely finite. The positive part is bounded by f^2. With $\gamma(x,v) = (2\pi)^{-d/2} e^{-|v|^2/2}$ and $c_\gamma = d\log(2\pi)/2$, the negative part on $\{f \le 1\}$ obeys

$$-f \log f \le \gamma/e + f(c_\gamma + |v|^2/2).$$

Integrating $r \log r - r + 1 \ge 0$ against γ, with $r = f/\gamma$, also gives

$$H(f) \ge -c_\gamma - \tfrac{1}{2} M(f). \tag{12}$$

3 The entropy comparison in the weak class

Fix $0 < \nu \le 1$. Let θ_ε be the centered Gaussian on $\mathbb{R}^d$ with covariance $\varepsilon^2 I_d$, and let ρ_ε be its periodization on $\mathbb{T}^d$. Define

$$R_\varepsilon f(t,x,v) = \int_{\mathbb{T}^d\times\mathbb{R}^d} \rho_\varepsilon(y)\theta_\varepsilon(w) f(t, x-y-tw, v-w)\,\mathrm{d}y\,\mathrm{d}w, \quad q_\varepsilon = R_\varepsilon m_\nu, \quad c_\varepsilon = R_\varepsilon a_\nu. \tag{13}$$

The shear tw makes this convolution commute with kinetic transport: for fixed (y,w),

$$D[f(t,x-y-tw,v-w)] = (Df)(t,x-y-tw,v-w).$$

It also commutes with velocity derivatives. Hence, distributionally,

$$Dq_\varepsilon - \nu\Delta_v q_\varepsilon + \operatorname{div}_v c_\varepsilon = 0. \tag{14}$$

The density is smooth and positive in (x,v) and has mass one. The translations have unit Jacobian, so

$$\mathcal{A}(q_\varepsilon, c_\varepsilon) \le \mathcal{A}(m_\nu, a_\nu), \qquad \mathcal{G}(q_\varepsilon) \le \mathcal{G}(m_\nu), \qquad M(q_\varepsilon(t)) = M(m_\nu(t)) + d\varepsilon^2. \tag{15}$$

Its initial density is $R_\varepsilon m_{\mathrm{in}}$ at $t=0$. We will recover m_{in} by taking $\varepsilon \to 0$ before using inviscid minimality.

We first justify the entropy identity for this positive density. In physical variables the convolution kernel is

$$\kappa_{\varepsilon,t}(x,v) = \rho_\varepsilon(x-tv)\theta_\varepsilon(v).$$

Weighted Cauchy–Schwarz in the periodizing sum gives $I(\rho_\varepsilon) := \int_{\mathbb{T}^d} |\nabla\rho_\varepsilon|^2/\rho_\varepsilon \le d/\varepsilon^2$. The velocity score of $\kappa_{\varepsilon,t}$ is $-t\nabla\log\rho_\varepsilon(x-tv) + \nabla\log\theta_\varepsilon(v)$. The integrated cross term is zero since $\int_{\mathbb{T}^d}\nabla\rho_\varepsilon = 0$. Applying the same weighted inequality to the convolution gradient gives

$$F_\varepsilon := \int_S \frac{|\nabla_v q_\varepsilon|^2}{q_\varepsilon} \le \frac{d(T+T^3/3)}{\varepsilon^2} < \infty. \tag{16}$$

For fixed ε, the kernel is bounded, so q_ε is uniformly bounded above. By (11), at least half the mass of $m_\nu(t)$ lies in one fixed velocity ball at every time. The positive periodic kernel ρ_ε has a positive minimum. Restricting the convolution to that ball therefore gives

$$q_\varepsilon(t,x,v) \ge c_{\varepsilon,*} e^{-C_\varepsilon(1+|v|^2)}, \qquad |\log q_\varepsilon(t,x,v)| \le C_\varepsilon(1+|v|^2), \tag{17}$$

with positive constants depending on ε. All kernel derivatives are bounded on $[0,T]$. On each compact velocity set, (14) thus puts $\partial_t q_\varepsilon$ in $L^1_t L^\infty_{x,v}$, using (7). The continuous representative furnished by convolution agrees with the weak representative, including at the endpoints. Positivity and the one-dimensional $W^{1,1}$ chain rule [4, Section 8.2] now justify differentiating $q_\varepsilon\log q_\varepsilon$ on such compact sets.

Let $\chi_R(v)$ equal one on $|v| \le R$, vanish on $|v| \ge 2R$, and satisfy $|\nabla\chi_R| \le C/R$, where $R \ge 1$. Multiply (14) by $\chi_R(1+\log q_\varepsilon)$ and integrate. The x-transport integrates to zero by periodicity. The velocity integration by parts produces the principal mixed and Fisher terms and the two boundary terms

$$\int_S (1+\log q_\varepsilon) c_\varepsilon\cdot\nabla\chi_R, \qquad -\nu\int_S (1+\log q_\varepsilon)\nabla_v q_\varepsilon\cdot\nabla\chi_R.$$

On the annulus supporting these terms, $(1+|v|^2)/R \le C(R^{-1}+|v|)$. Both terms tend to zero because

$$\int_S |v||c_\varepsilon| \le \sqrt{2\mathcal{A}(q_\varepsilon, c_\varepsilon)} \left(\int_S |v|^2 q_\varepsilon\right)^{1/2},$$
$$\int_S |v||\nabla_v q_\varepsilon| \le \sqrt{F_\varepsilon} \left(\int_S |v|^2 q_\varepsilon\right)^{1/2}$$

are finite; the corresponding unweighted integrals are finite as well. The principal mixed term is absolutely integrable by Cauchy–Schwarz and (16). Endpoint entropies pass by absolute integrability, as established above. We obtain

$$H(q_\varepsilon(T)) - H(q_\varepsilon(0)) = \int_S c_\varepsilon \cdot \nabla_v \log q_\varepsilon - \nu F_\varepsilon. \tag{18}$$

Only the smoothed density has been used in this chain rule.

Define the current flux $b_\varepsilon = c_\varepsilon - \nu \nabla_v q_\varepsilon$. Then $Dq_\varepsilon + \operatorname{div}_v b_\varepsilon = 0$. Its action is finite by (15) and (16). Expanding the square and using (18) gives

$$\mathcal{A}(q_\varepsilon, b_\varepsilon) = \mathcal{A}(q_\varepsilon, c_\varepsilon) - \nu[H(q_\varepsilon(T)) - H(q_\varepsilon(0))] - \frac{\nu^2}{2} F_\varepsilon. \tag{19}$$

Initial convolution decreases entropy by convexity of $s \log s$. The terminal entropy is bounded below by (12) and (15). For $0 < \varepsilon \le 1$, these give

$$H(q_\varepsilon(0)) - H(q_\varepsilon(T)) \le H_{\text{in}} + c_\gamma + (K+d)/2 \le L.$$

Keeping the negative Fisher term in (19) until this point, and then discarding it, yields

$$\mathcal{J}(q_\varepsilon, b_\varepsilon) \le E_\nu + L\nu \qquad (0 < \varepsilon \le 1), \tag{20}$$

with the constant L in (5). This estimate is uniform in smoothing, although (16) itself is not.

4 Removing the convolution and proving the rate

We remove smoothing with ν fixed. Translation continuity and the Gaussian shift moments in (13) imply

$$\begin{aligned} q_\varepsilon &\longrightarrow m_\nu && \text{in } L^1(S) \cap L^2(S), \\ q_\varepsilon(T) &\longrightarrow m_\nu(T) && \text{in } L^2(Q), \\ q_\varepsilon(0) &\longrightarrow m_{\text{in}} && \text{in } L^1(Q). \end{aligned} \tag{21}$$

For spacetime convergence, apply translation continuity at almost every time and dominated convergence to the contracted L^p norms. The bounded shear on $[0,T]$ makes the shift size tend to zero uniformly in probability.

Hölder's inequality gives the flux bound

$$\int_S |b_\varepsilon|^{4/3} \le \left(\int_S \frac{|b_\varepsilon|^2}{q_\varepsilon}\right)^{2/3} \left(\int_S q_\varepsilon^2\right)^{1/3}. \tag{22}$$

Both factors are uniformly bounded by (20) and (15). Choose a sequence $\varepsilon_n \downarrow 0$ realizing the lower action limit A_*. Reflexivity and weak sequential compactness in $L^{4/3}(S;\mathbb{R}^d)$ [4, Theorems 3.18

and 4.10] give, along a subsequence, $b_{\varepsilon_n} \rightharpoonup b$. These spaces are defined on the full noncompact domain. For example, the velocity tails also satisfy

$$\int_{\{|v|>R\}} |b_\varepsilon| \le \sqrt{2\mathcal{A}(q_\varepsilon, b_\varepsilon)}\sqrt{T(K+d)/R^2} \qquad (0<\varepsilon\le 1).$$

We check that the limiting flux has finite action with precisely the original vacuum convention. For every compactly supported continuous vector field z on S, completing a square and passing to the limit give

$$\Phi(z) - \frac{1}{2}\|z\|^2_{L^2(\mu)} \le A_*, \qquad \Phi(z) = \int_S z\cdot b, \qquad \mathrm{d}\mu = m_\nu \,\mathrm{d}t\,\mathrm{d}x\,\mathrm{d}v. \tag{23}$$

Replacing z by each real multiple of itself shows $|\Phi(z)| \le \sqrt{2A_*}\|z\|_{L^2(\mu)}$. Continuous compactly supported fields are dense in this weighted space. To see this without a lower bound on m_ν, first truncate a weighted L^2 field in size and support. Approximate the resulting bounded field in Lebesgue L^2 by continuous compactly supported fields and clip the approximants to a common size bound. On $\{m_\nu \le R\}$ the weighted error is at most R times the Lebesgue squared error; the remaining error is bounded by a constant times $\int_{\{m_\nu>R\}} m_\nu$, which tends to zero. Lebesgue density on the torus follows by approximation in a fundamental cell and periodization; the Euclidean density result is [4, Theorem 4.12].

Consequently Φ extends to the Hilbert space $L^2(\mu;\mathbb{R}^d)$. The Riesz representation theorem [4, Theorem 5.5] provides u in that space with norm at most $\sqrt{2A_*}$ and $\Phi(z) = \int_S z\cdot u\,\mathrm{d}\mu$. Both b and um_ν are locally integrable; equality against compact smooth tests identifies them almost everywhere. Hence $b = um_\nu$, including $b=0$ on $\{m_\nu = 0\}$, and

$$\mathcal{A}(m_\nu, b) \le A_* = \liminf_n \mathcal{A}(q_{\varepsilon_n}, b_{\varepsilon_n}). \tag{24}$$

In particular, b is integrable by (7).

Now pass to the inviscid weak equation for $(q_{\varepsilon_n}, b_{\varepsilon_n})$. The gradient of any smooth compact-velocity test belongs to $L^4(S)$, so the flux pairing passes under the weak convergence above, including for tests reaching a time endpoint. All density and endpoint terms pass by (21). The result is (3) with $\nu = 0$, initial density exactly m_{in}, and density curve m_ν. The narrow continuity, probability property, and finite moments are those of this original curve. Strong convergence of both square costs and (24) therefore show that (m_ν, b) is an inviscid admissible pair with

$$\mathcal{J}(m_\nu, b) \le E_\nu + L\nu \le E_0 + L\nu. \tag{25}$$

The construction also identifies $b = a_\nu - \nu\nabla_v m_\nu$ distributionally. No differentiability of m_ν was assumed to construct it.

Proof of Theorem 1. The midpoint of (m_0, a_0) and (m_ν, b) is admissible for zero diffusivity: the weak equation is linear, the initial traces agree, and the probability and moment conditions are preserved. Perspective convexity and the exact square identity in both density costs give

$$\begin{aligned} E_0 &\le \mathcal{J}\left(\frac{m_0+m_\nu}{2}, \frac{a_0+b}{2}\right) \\ &\le \frac{E_0 + \mathcal{J}(m_\nu, b)}{2} - \frac{1}{8}\|m_\nu - m_0\|^2_{L^2(S)} - \frac{1}{8}\|m_\nu(T) - m_0(T)\|^2_{L^2(Q)}. \end{aligned}$$

Insert (25) and rearrange to obtain (4). In particular, for every stipulated minimizing pair and every $0<\nu\le 1$,

$$\|m_\nu - m_0\|_{L^2((0,T)\times\mathbb{T}^d\times\mathbb{R}^d)} \le 2\sqrt{L}\,\nu^{1/2}.$$

Here L is given by (5): it depends only on d, T, the initial L^2 norm, second velocity moment, and entropy. All are finite for the stated datum, and none depends on ν. Thus the exponent $\eta(d) = 1/2$ applies in every dimension. □

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References

[1] W. Tang and Y. P. Zhang, *The convergence rate of vanishing viscosity approximations for mean field games*, SIAM Journal on Mathematical Analysis **57** (2025), 3217–3254. doi:10.1137/24M1640008.

[2] M. Griffin-Pickering and A. R. Mészáros, *A variational approach to first order kinetic mean field games with local couplings*, arXiv:2112.03141v3, 11 July 2022.

[3] J.-D. Benamou, G. Carlier, S. Di Marino, and L. Nenna, *An entropy minimization approach to second-order variational mean-field games*, arXiv:1807.09078v2, 29 March 2019.

[4] H. Brezis, *Functional Analysis, Sobolev Spaces and Partial Differential Equations*, Universitext, Springer, 2011. doi:10.1007/978-0-387-70914-7.